Yandex

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Switchdev in the Wild

Anton Kortunov, Tech Lead of nocdev team

Yandex Network Overview

We’re hyperscale cloud operator

- Commodity hardware + scale out
- Pooled resources
- Cluster scale “operation system”
- We control endpoints, network and software stack

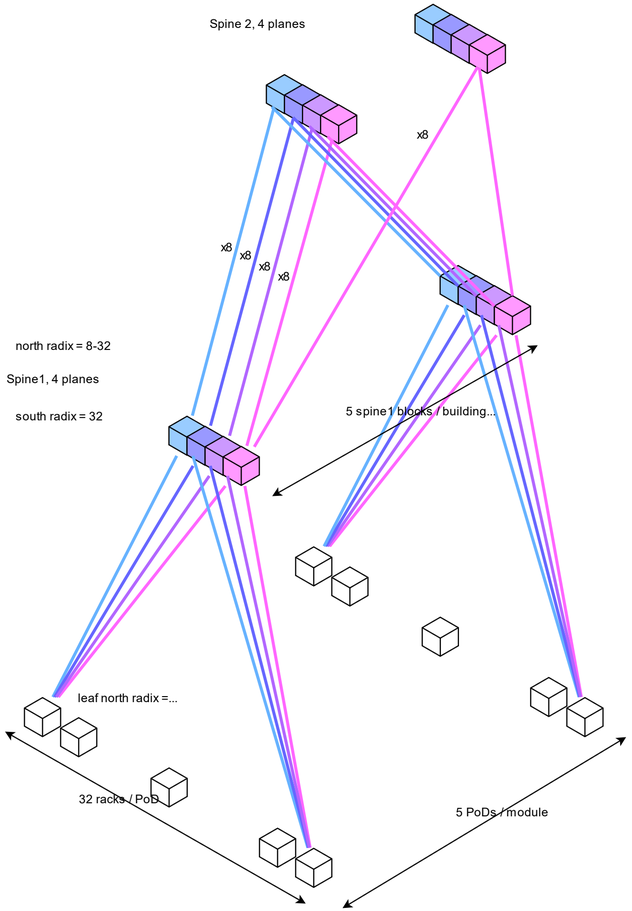
Several large DCs in Russia and abroad + MPLS backbone DC interconnect

Recent DCN:

- Multi-stage CLOS
- IPv6-only fabric
- L3 ToR

Up to 100k servers per DC

DCN Simplified Topology



ToR Required SW Features

- BGP on IPv6 link-local addresses
- QoS
- ACL counters
- Telemetry and monitoring
- Automation of config deployment

Linux-based NOSes We Tried

- **Cumulus**
- **SONiC**
- **Switchdev**

Switchdev

Thank you, Alexander Petrovskiy!

Based on Ubuntu Bionic Beaver
Systemd as an init system
Mellanox chassis management scripts
Ifupdown2 from Cumulus
VRF from Alexander

Switchdev Configuration Automation

linux-commit-api based on etckeeper

- Git as a storage backend
- Resolves systemd services
- Allows to rollback changes
- Requires services to be idempotent

Tacplus for authentication

- Patched sudo
- Tacacs server with centralized access control

Switchdev Deployment Automation

ONIE stage

- Lack of native IPv6 support
- Lack of data in DHCP requests

ZTP in Linux

- No common implementation
- Based on dhclient
- Need way to transfer deployment logs
- Inconsistent with our deployment infrastructure (console-based)

Devlink subsystem

Powerful tool

- Portsplit configuration
- Traps and policers
- Resources usage

Hard to use from code

- Lot of thigs done under the hood of `devlink` program itself
- We use as a tool from bash scripts
- Hard to write idempotent scripts

TC

Even more powerful tool!

- QoS
- Queue management
- ACL
- Counters
- Specific features: mirroring, sampling, vlan modify

Hard to use from code

Even hard to understand

- Not intuitive for NOC guys

Future Plans

gNMI and gNOI

- Uniform access to different counters: `ethtool`, `tc`, `ip link`, etc
- Simplify automation tools

Decrease number of configs

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Thank you

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